Momentum & Trend-Following Strategy

A Systematic Approach to Risk-Adjusted Investing on the S&P 500

Fitte-Duval Gaëlle
Quantitative Finance Research · March 2026

ABSTRACT

Can a simple set of rules — applied consistently, without emotion — protect investors from devastating market crashes while still participating in long-term growth? This paper tests exactly that. We design a systematic trading strategy on the S&P 500 ETF (SPY) over 20 years (2005–2024), combining three clear signals: a long-term trend filter, a momentum check, and an overbought guard. The result: a strategy that cuts the worst losses by more than half — a maximum drop of 17% versus 42% for simply holding the market — while delivering solid risk-adjusted returns. Transaction costs, realistic assumptions, and critical limitations are discussed with full transparency.

0.54	–17%	0.53	9.1%	9.4%
Sharpe Ratio	**Max Drawdown**	**Calmar Ratio**	**Annual Return**	**Ann. Volatility**
risk-adjusted	*vs –42% B&H;*	*return/risk*	*vs 17% B&H;*	*vs 19.6% B&H;*

CONTENTS

1. Introduction . . .
2. Data & Methodology . . .
3. The Strategy Explained . . .
4. Volatility & Options Pricing . . .
5. Performance Results . . .
6. Limitations & Risks . . .
7. Conclusion . . .
References . . .

Simulation based on synthetic regime-calibrated data. Past performance is not indicative of future results.

© Fitte-Duval Gaëlle · 2026

1. INTRODUCTION

Most investors face a painful dilemma: stay invested and risk devastating losses during market crashes, or sit in cash and miss the rally. This strategy offers a third path — a rule-based system that automatically reduces exposure when markets become dangerous, and increases it when conditions are favourable.

The academic literature has long documented two reliable market patterns. **Momentum**: assets that have risen recently tend to keep rising for a few months (Jegadeesh & Titman, 1993). **Trend-following**: being out of the market during sustained downtrends dramatically reduces drawdowns (Moskowitz, Ooi & Pedersen, 2012). This strategy combines both.

We test the strategy on SPY — the world's most traded exchange-traded fund, tracking the S&P 500 — over 20 years (2005–2024). This period includes every major crisis of the modern era: the 2008 financial collapse, the 2020 COVID crash, and the 2022 rate-hiking bear market.

2. DATA & METHODOLOGY

2.1 Where the Data Comes From

Daily adjusted closing prices for SPY are sourced from Stooq.com, covering approximately 5,040 trading days. For this backtest, synthetic regime-calibrated data is used — price series constructed to faithfully replicate the volatility, crashes, and rallies of actual market history.

We compute the daily return — the percentage change from one day to the next — as our core measurement. This tells us how much the investment gained or lost each day.

```
Daily Return = (Today's Price – Yesterday's
        Price) / Yesterday's Price
```

2.2 The Historical Record

The dataset spans three complete bull markets (sustained rises) and two severe bear markets (peak-to-trough drops exceeding 30%). This range of conditions is essential: a strategy that only works in one type of market is not truly robust.

3. THE STRATEGY EXPLAINED

3.1 Three Simple Rules

Before investing any money, the strategy checks three conditions. **All three must be true simultaneously** for the strategy to invest:

ENTRY SIGNAL — invest only when:

① The 50-day average price is above the 200-day average *(the market is in an uptrend)*

② The price is higher than 3 months ago *(recent momentum is positive)*

③ The RSI indicator is below 70 *(the market is not overheating)*

The **50/200-day moving average crossover** is one of the oldest and most studied signals in finance. When the short-term average rises above the long-term average, the market is considered to be in a healthy uptrend. The **RSI (Relative Strength Index)** measures whether buying has become excessive — a reading above 70 historically signals a short-term pullback risk.

3.2 How Much to Invest: Volatility Targeting

Once the three conditions are met, the strategy does not simply invest 100% of the portfolio. Instead, it adjusts the investment size based on how volatile — how turbulent — the market is currently.

When volatility is high (as during the 2008 crash or March 2020): the strategy automatically reduces its position. **When volatility is low** (calm, trending markets): it invests more fully. The target is always a steady 12% annual volatility in the portfolio — regardless of what the market is doing.

```
Position Size = clip( 12% Target / Current
        Volatility, 30%, 150% )
```

This is the single most important feature of the strategy. It is the reason the maximum loss was –17% while the market fell –42%: the strategy held much less during the worst periods.

3.3 Transaction Costs

Every time the strategy changes its position, it incurs a cost — the difference between the buying and selling price (the bid-ask spread). We apply a realistic cost of 5 basis points (0.05%) per change. For SPY — the most liquid ETF in the world — this is conservative but honest.

© Fitte-Duval Gaëlle · 2026

4. VOLATILITY & OPTIONS

4.1 Measuring Market Volatility (EWMA)

To size the position correctly, we need a real-time estimate of market volatility. We use the EWMA method (Exponentially Weighted Moving Average), developed by J.P. Morgan's RiskMetrics team in 1994. It gives more importance to recent days and less to older ones — capturing how quickly volatility can spike during a crisis.

```
Tomorrow's Vol² = 0.94 × Today's Vol² + 0.06 × Today's Return²
```

The parameter 0.94 means roughly half the 'weight' sits in the last 12 trading days — two and a half calendar weeks. This makes the estimate very responsive to sudden market moves.

4.2 Pricing an Options Contract (Black-Scholes)

Using our volatility estimate, we can price a standard financial instrument: a European call option on SPY. A **call option** gives the right (but not the obligation) to buy SPY at a fixed price on a future date. The Black-Scholes formula (1973) is the industry-standard pricing model:

```
Call Price = S × N(d1) – K × e^(–rT) × N(d2)
```

With our estimated volatility of 12.31% at end of 2024, the at-the-money 3-month call option is valued at **$136.85**.

Greek	Value	Plain English
Delta Δ	0.577	If SPY rises $1, option gains $0.58
Gamma Γ	0.00138	Delta accelerates as price moves
Vega V	$9.01	Option gains $9 per +1% volatility
Theta Θ	–$/day	Option loses value each passing day

5. PERFORMANCE RESULTS

5.1 The Numbers

Table 1. Backtest results — SPY, January 2005 to December 2024

Metric	Strategy	Buy & Hold
Annual Return (%)	9.08	**17.00**
Annual Volatility (%)	**9.42**	19.59
Sharpe Ratio	0.539	**0.663**
Sortino Ratio	0.679	**0.999**
Max Drawdown (%)	**–17.09**	–41.82
Calmar Ratio	**0.531**	0.406
Win Rate (%)	31.73	**52.98**
Annual Turnover	17.24	**0.00**

Green = strategy advantage · Blue = comparable · Red = buy-and-hold advantage

5.2 What Do These Numbers Mean?

The strategy earns less over time — but that is entirely by design. Buy-and-hold earns 17% per year because it is invested 100% of the time, including through catastrophic crashes it cannot avoid. The strategy earns 9.1% because it steps aside during danger — and that selective investing is what limits losses.

The key advantages of the strategy are:

— **Maximum loss cut by 60%**: worst drawdown of –17% versus –42% for the market. A –42% loss requires a +72% gain just to break even.

— **Calmar ratio 31% higher** (0.53 vs 0.41): more return earned per unit of maximum loss suffered.

— **Volatility halved**: 9.4% versus 19.6% — the portfolio is far smoother and less stressful to hold.

5.3 Monte Carlo: What Could Happen Next Year?

Running 1,000 simulated futures based on the strategy's historical return patterns:

— Median outcome (most likely): +9% gain

— Worst 5% of scenarios: –12% loss

— Best 5% of scenarios: +30% gain

6. LIMITATIONS & RISKS

Intellectual honesty requires stating clearly what this backtest cannot guarantee:

1. **Overfitting risk.** The moving average crossover (50/200 days) has been studied so extensively that it may have been unconsciously tuned to past data. Always test on data the model has never seen.
2. **Survivorship bias.** SPY reflects the success of the U.S. market — one of history's best performers. The same strategy applied to other markets might show different results.
3. **Cost underestimation.** 5 basis points is realistic for small portfolios today. For larger amounts, or in 2005 market conditions, costs were higher.
4. **Fat tails.** Extreme events (crashes, flash crashes) happen more often than standard statistical models predict. Risk models that assume normal distributions will underestimate tail risk.
5. **Regime dependence.** The strategy underperforms in prolonged sideways markets with no clear trend. It is designed for trending environments.

7. CONCLUSION

A simple, rule-based strategy — applied with discipline over 20 years — can meaningfully improve the experience of investing. Not by earning more than the market every year, but by avoiding the worst crashes that destroy wealth and test investor resolve.

© Fitte-Duval Gaëlle · 2026

The momentum and trend-following strategy demonstrated here reduced the worst loss from –42% to –17%, halved portfolio volatility, and improved the return-to-risk ratio — all while remaining fully systematic, transparent, and reproducible.

Full source code: Python (pandas, numpy, scipy, matplotlib)

github.com/[username]/quant-strategy-spy

REFERENCES

[1] Jegadeesh, N. & Titman, S. (1993). Returns to buying winners and selling losers. *Journal of Finance*, 48(1), 65–91.

[2] Moskowitz, T., Ooi, Y.H. & Pedersen, L.H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.

[3] Black, F. & Scholes, M. (1973). Pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.

[4] J.P. Morgan / Reuters (1996). *RiskMetrics Technical Document*, 4th ed.

[5] Fama, E. (1970). Efficient capital markets. *Journal of Finance*, 25(2), 383–417.

© Fitte-Duval Gaëlle · 2026

FIGURE 1 — PERFORMANCE DASHBOARD

Cumulative Wealth	How $1 invested in 2005 grew over time. The strategy (teal line) grows more slowly but avoids the severe crashes of simply holding the market (red line).
Drawdown Profile	The percentage fall from each peak. The strategy's worst fall was –17% (teal). The market fell –42% in 2008 and again in 2022 (red).
Rolling Sharpe Ratio	A measure of risk-adjusted performance over each 12-month window. Above zero = the strategy is earning more than its volatility warrants.
Monte Carlo	1,000 possible futures simulated from the strategy's return history. The shaded bands show the range of likely outcomes over the next year.

Momentum Strategy (teal) vs Buy & Hold SPY (red) · Simulation based on regime-calibrated synthetic data · 2005–2024

© Fitte-Duval Gaëlle · 2026